Bash Scripting. Automation

Bash Scripting Building Blocks

Repetitive Tasks Automation

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THIS MODULE

AND ONE MORE TO GO

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** It is hosted externally, and it is not part of SoftUni's infrastructure. Requires registration (sign up)*

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There Should be an Exam Sign-up Form

** It could appear even earlier. There will be a message in the Facebook group when the sign-up form is available*

Previous Module (M6)

Quick Overview

What We Covered

- Filesystem Hierarchy Standard (FHS)
- Archiving Tools
- Disks and Partitions Schemes
- File Systems

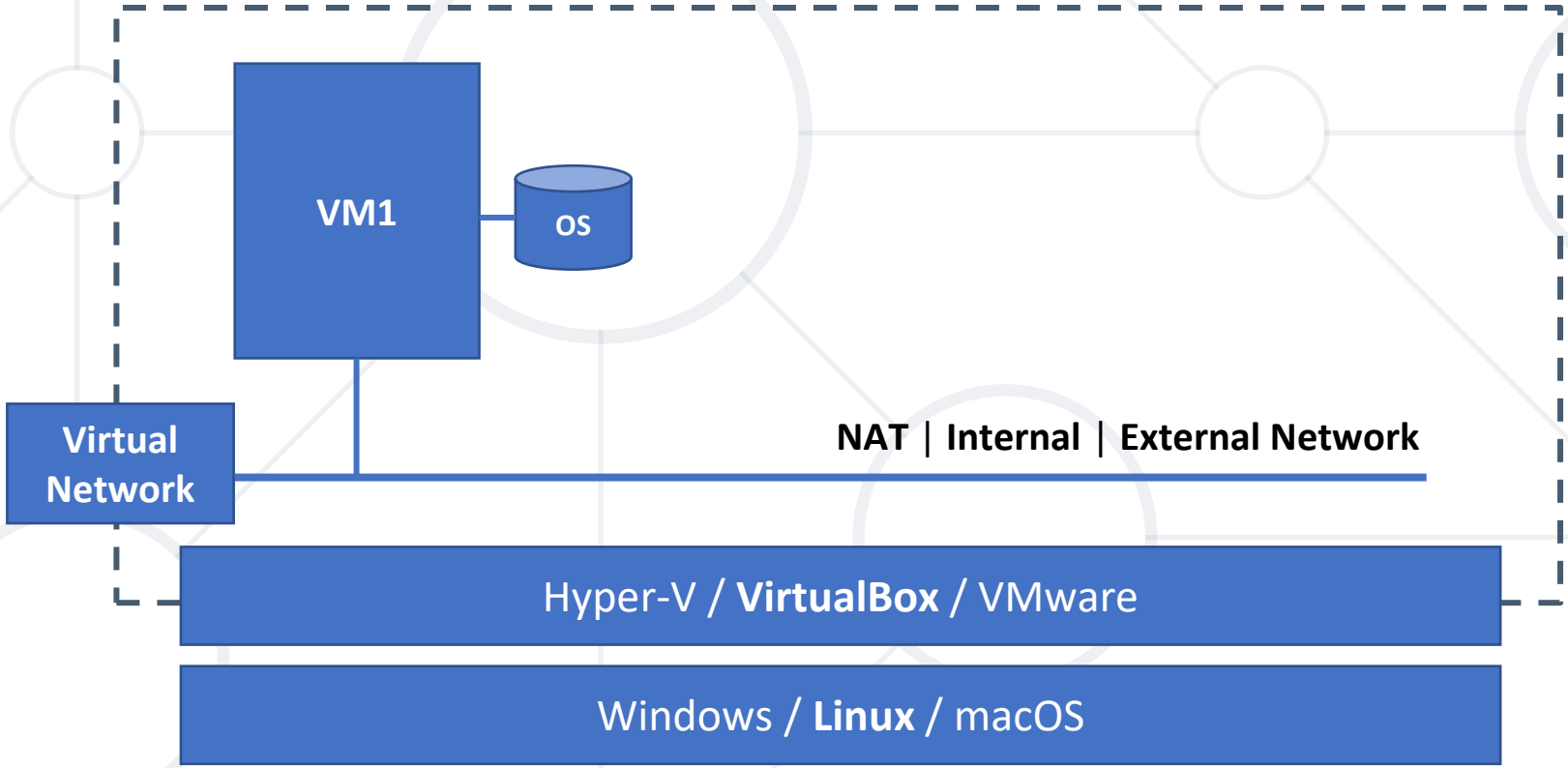
This Module (M7)

Topics and Lab Infrastructure

Table of Contents

1. Scheduled task execution
2. Bash scripts building blocks
3. Writing scripts in bash

Lab Infrastructure



Scheduling

Periodical Task Execution

Purposes

- Regular and repetitive tasks
  - Cleaning, archiving, monitoring, …
- Runtime varies
  - Schedule based or one-time, but at specific moment
- Defined on system or user level

cron Introduction

- **cron** is the main task scheduler in Linux
- Components
  - **crond** => Daemon
  - **crontab** => Management tool
- Configuration files
  - **Tasks** are read from **/etc/crontab** and **/etc/cron.d/***
  - **Rights** are read from **/etc/cron.allow** and **/etc/cron.deny**
- Per user jobs
  - **/var/spool/cron/***

cron Format

```
* * * * * user-name /path/to/executable
```

- Command to execute
- User to be used for the job *
- Day of Week (0-7) or **sun**, **mon**, … **
- Month (1-12) or **jan**, **feb**, **mar**, …
- Day of Month (1-31)
- Hour (0-23)
- Minute (0-59)

* User is **not specified** in the per user crontabs (**/var/spool/cron/user**)

** **Sunday** can be set either with **0** or **7**

cron Examples

```
# Execute every minute
* * * * * /utilities/backup.sh
# Run every noon at 12:00
0 12 * * * /utilities/backup.sh
# Run on 1st of January at 00:00
0 0 1 1 * /utilities/backup.sh
# Run every Monday at 5:30
30 5 * * 1 /utilities/backup.sh
# Run at 00:00 and at 12:00 every day
0 0,12 * * * /utilities/backup.sh
# Run every two hours every day
0 */2 * * * /utilities/backup.sh
# Run hourly between 9 and 17 o'clock every day
0 9-17 * * * /utilities/backup.sh
```

cron Shortcuts

- **@yearly or @annually**
  - Run once a year at midnight of 1st of January (**0 0 1 1 ***)
- **@monthly**
  - Run once a month at midnight of the first day (**0 0 1 * ***)
- **@weekly**
  - Run once a week at midnight on Sunday morning (**0 0 * * 0**)
- **@daily or @midnight**
  - Run once a day at midnight (**0 0 * * ***)
- **@hourly**
  - Run once an hour at the beginning of the hour (**0 * * * ***)

anacron

- Runs commands periodically with **frequency in days**
- It does not assume that the machine is non-stop powered
- For each job **anacron** checks if it has been executed in the last **N** days, where **N** is the interval specified for the job
- Jobs are stored at **/etc/anacrontab**
- Configuration can be tested with **anacron -T**
- Shortcuts – **@daily** or **1**, **@weekly** or **7**, and **@monthly** or **30**

at

- Run a task once at a specific time (you may need to install **at** package)
- Each task is queued at **/var/spool/at**
- Security is defined through **/etc/at.allow** and **/etc/at.deny**
- Tools
  - **at** => Main utility
  - **batch** => Auxiliary utility can be used as at to schedule commands
  - **atq** => Show jobs at **at**'s queue
  - **atrm** => Delete at jobs
- Shortcuts – **today**, **midnight**, **noon**, **teatime**, **date**, **now + time unit**

systemd timer

- Systemd unit files (**.timer**) that control services (**.service**)
- Read from the **same paths** as the other units
- Offer built-in support for **calendar** and **monotonic** events
- Calendar (realtime) timers work the same way as cron jobs
- Monotonic timers activate after a time span relative to a point
- Can be created as **transient** (temporary/on the fly) units as well
- Can be used as an **alternative to cron**

systemd (calendar) timer

/etc/systemd/system/free-mem.timer

```
[Unit]
Description=Runs a service every
day at 04:00

[Timer]
OnCalendar=*-*-* 4:00:00
Persistent=true

[Install]
WantedBy=timers.target
```

/etc/systemd/system/free-mem.service

```
[Unit]
Description=Logs system free
memory
Wants=free-mem.timer

[Service]
Type=oneshot
ExecStart=/usr/bin/free

[Install]
WantedBy=multi-user.target
```

1) We can have more than one **OnCalendar** item

2) **Persistent=true** enables immediate execution after activation if it missed the last start time (if the system was off)

https://www.freedesktop.org/software/systemd/man/systemd.timer.html

OnCalendar

- Has the following format
  - **DayOfWeek Year-Month-Day Hour:Minute:Second**
- DayOfWeek can be specified as **Mon**, **Monday**, **mon**, or **monday**
- There are some special expressions, for example:
  - **monthly** -> *-*-01 00:00:00
  - **weekly** -> Mon *-*-* 00:00:00)
- We can test expressions with **systemd-analyze calendar**

https://www.freedesktop.org/software/systemd/man/systemd.time.html

systemd (monotonic) timer

/etc/systemd/system/free-mem.timer

```
[Unit]
Description=Runs weekly and on
boot

[Timer]
OnBootSec=10min
OnUnitActiveSec=1w

[Install]
WantedBy=timers.target
```

/etc/systemd/system/free-mem.service

```
[Unit]
Description=Logs system free
memory
Wants=free-mem.timer

[Service]
Type=oneshot
ExecStart=/usr/bin/free

[Install]
WantedBy=multi-user.target
```

1) Other options are **OnActiveSec**, **OnStartupSec**, and **OnUnitInactiveSec**

2) Expressions can be tested with **systemd-analyze timespan**

https://www.freedesktop.org/software/systemd/man/systemd.timer.html

systemd (transient) timer

- Starting an arbitrary command

```
systemd-run --on-active=30 /bin/touch /tmp/file
```

- Starting an existing service unit

```
systemd-run --on-active="6h 15m" --unit free-mem.service
```

https://wiki.archlinux.org/title/Systemd/Timers

Practice

Scripts

Structure and Variables

Structure

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Script Signature

Comments

Local Variable

Command

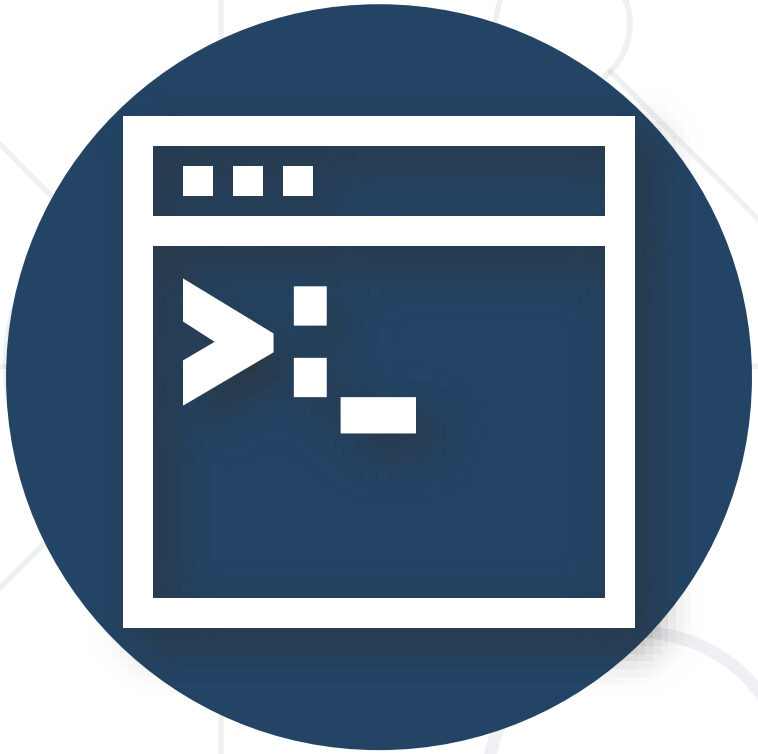
```
#!/bin/bash

#
# hello.sh
#

attn="dear"

echo "Hello $attn $USER"
```

- **Execution**: `bash hello.sh` or `./hello.sh` or just `hello.sh`



Commands and Flow Control #1

echo

- Description
  - Display line of text
- Example

```
[user@host ~]$ echo 'Hello world!'
Hello world!


[user@host ~]$ echo 'Current user: '$USER
Current user: user
```

printf

- Description
  - Formats and prints text
- Example

```
[user@host ~]$ printf 'Hello world!\n'
Hello world!


[user@host ~]$ printf 'I say %d is the answer\n' 42
I say 42 is the answer
```

seq

- Description
  - Count from starting to ending point
- Example

```
$ seq 1 5
1 2 3 4 5
$ seq 1 2 5
1 3 5
$ seq -w 5 10
05 06 07 08 09 10
```

for (1)

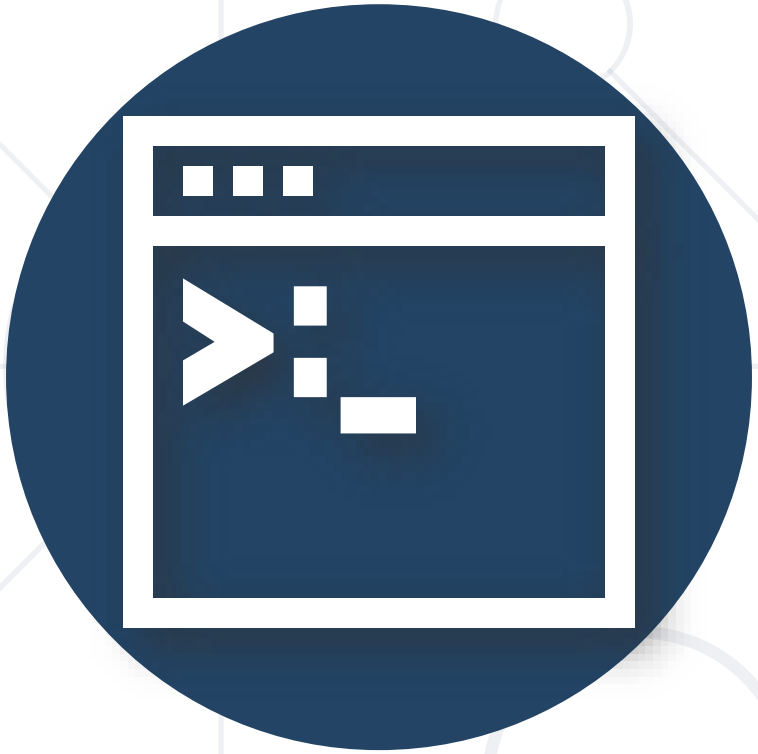
- Description
  - Execute command for each member in a list
- Example

```
# List all files with prefix “item:”
for i in $( ls ); do
   echo item: $i
done
# Create files fileXX.txt where XX is between 05 and 10
for i in $( seq -w 5 10 ); do
   touch file$i.txt
done
```

for (2)

```
# Iterate over the elements of a list
for i in {1..10}; do
   echo item: $i
done
# C-style for loop
for ((i=1;i<=10;i++)); do
   echo item: $i
done
# Nested loops
for i in {1..10}; do
   for j in {1..10}; do
      echo $i-$j
   done
done
```

Practice



Commands and Flow Control #2

test

- Description
  - Evaluate conditional expression
- Example

```
# Compare numbers: OP1 -eq|-ne|-lt|-le|-gt|-ge OP2

# Compare strings: ST1 =|!=|<|> ST2

# Compare files: FL1 -nt|-ot FL2

# File tests: -d|-e|-f|-x FILE
```

if

- Description
  - Execute commands based on conditional
- Example

```
count=1
if [ $count -eq 0 ]; then
  echo 'Equal to 0'
else
  echo 'Not equal to 0'
fi
```

while

- Description
  - Execute commands as long as a test succeeds
- Example

```
# Print numbers from 1 to 5
count=1
while [ $count -le 5 ]; do
  echo $count
  count=$((count+1))
done
```

until

- Description
  - Execute commands as long as a test does not succeed
- Example

```
# Print numbers from 1 to 5
count=1
until [ $count -gt 5 ]; do
  echo $count
  count=$((count+1))
done
```

case

- Description
  - Execute commands based on conditional
- Example

```
count=1
case $count
 1) echo 'One'
    ;;
 2|3|4) echo 'Two, Three, or Four'
    ;;
 *) echo 'Something Else'
esac
```

Scripts with Parameters and Prompts

Special Variables

- Name of the script **$0**
- Positional arguments **$1** .. **$9**, **${10}**, **${11}** …
- Total number of arguments **$#**
- List of positional parameters **$*** or **$@**
- Exit code of last executed command **$?**

read

- Description
  - Read a line from the standard input and split it into fields
- Example

```
[user@host ~]$ read -p "Enter name:" NM_ENT
Enter name: James


[user@host ~]$ echo $NM_ENT
James
```

Work with Prompt

- Interactive prompt for user input

```
#!/bin/bash

# Ask for user input

read -p 'Enter your name: ' USR_NAME

echo 'Hello, '$USR_NAME
```

Accept One Parameter

- Check and accept just one parameter

```
#!/bin/bash

# Accept one parameter

if [ $# -ne 1 ]; then
  echo 'Usage: '$0' your_name';
  exit 1;
fi

echo 'Hello, '$1
```

Practice

Summary

- Sourcing is an alternative approach to script execution
- Sourcing can be done in two ways source script.sh or . script.sh
- cron and at are tools for scheduling tasks execution
- Systemd times can be used to schedule tasks as well

Summary

- Bash scripts are built from comments (#) and commands
- Bash scripts can accept parameters on the command line and user input
- We can use flow-control (if, case) and loop (for, while, until) commands

Resources

- Bash Programming – Introduction How-To
  - http://tldp.org/HOWTO/Bash-Prog-Intro-HOWTO.html
- Bash Reference Manual
  - https://www.gnu.org/software/bash/manual/html_node/index.html
- Cron How-To
  - https://help.ubuntu.com/community/CronHowto
- Cron Schedule Expressions Editor
  - https://crontab.guru/

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